

# IEC 62368-1 Standard Compliance Guide



We use cookies to improve your experience on our site. By using our site, you consent to cookies.

[Preferences](#)
[Reject](#)
[Accept All](#)
[WPConsent](#)

IEC 62368-1 is the international safety standard for audio/video, information technology, and communications equipment, replacing IEC 60950-1 and IEC 60065 across most global markets. This guide provides manufacturers with the essential compliance requirements, testing processes, and implementation strategies needed to achieve certification under this hazard-based safety engineering standard.

## What you'll learn in this guide:

- The fundamental principles of Hazard-Based Safety Engineering (HBSE) and how it differs from prescriptive standards
- Energy source classifications and required safeguards for compliance
- Key compliance requirements across electrical, thermal, mechanical, and fire safety
- Global market transition timelines and certification pathways
- How to streamline your testing and certification process



## Understanding Hazard-Based Safety Engineering (HBSE)

IEC 62368-1 represents a fundamental shift from traditional incident-based safety standards to a performance-oriented, hazard-based approach. Rather than prescribing specific construction methods based on past incidents, HBSE identifies potential energy sources within equipment and classifies them according to their potential to cause pain or injury.

### The Four-Step HBSE Methodology

The standard applies a systematic four-step process to evaluate product safety:

We use cookies to improve your experience on our site. By using our site, you consent to cookies.

[Preferences](#)
[Reject](#)
[Accept All](#)
[WPConsent](#)

This methodology allows manufacturers greater flexibility in product design while maintaining safety, as multiple safeguard approaches can achieve the same level of protection.

## Energy Source Classifications and User Types

IEC 62368-1 establishes a hazard-based framework that defines both energy source classifications and user types to determine appropriate safeguard requirements.

### Energy Source Classes

IEC 62368-1 classifies all energy sources into three distinct classes based on their potential to cause harm:

| Energy Class   | Characteristics                                             | Safeguard Requirements                                    |
|----------------|-------------------------------------------------------------|-----------------------------------------------------------|
| <b>Class 1</b> | Not painful, may be detectable; ignition unlikely           | No safeguard required for ordinary persons                |
| <b>Class 2</b> | Painful but not injurious; ignition possible                | At least one safeguard is required for ordinary persons   |
| <b>Class 3</b> | Capable of causing serious injury or death; ignition likely | At least two safeguards are required for ordinary persons |





## User Classifications

The standard recognizes three user types, each requiring different levels of protection:

| User Type         | Definition                                                        | Safeguard Level                                         |
|-------------------|-------------------------------------------------------------------|---------------------------------------------------------|
| Ordinary Person   | General public with no specialized knowledge                      | Highest protection required                             |
| Instructed Person | An individual with proper authorization and basic safety training | Moderate protection; may access Class 2 sources         |
| Skilled Person    | Technically qualified professional                                | Minimal safeguards; may work with higher energy classes |

## Core Compliance Requirements

Core compliance under IEC 62368-1 extends beyond electrical design to include fire enclosure performance and protection against mechanical hazards.

Electrical safety forms the foundation of IEC 62368-1 compliance. Manufacturers must evaluate:

- **Primary circuit protection:** Overcurrent and overvoltage safeguards
- **Secondary circuit isolation:** Proper separation between primary and secondary circuits
- **Grounding and bonding:** Adequate protective earth connections where required
- **Insulation requirements:** Appropriate working voltages and pollution degrees
- **Accessible parts:** Limitation of voltage on user-accessible surfaces

### Thermal and Fire Safety

Fire enclosure requirements depend on the energy sources present and their classification:

| Fire Enclosure Type | Application                            | Test Requirements       |
|---------------------|----------------------------------------|-------------------------|
| Type A              | Encloses Class 3 fire ignition sources | Glow wire test at 960°C |
| Type B              | Encloses Class 2 fire ignition sources | Glow wire test at 850°C |
| Type C              | Class 1 fire ignition sources only     | Glow wire test at 750°C |

Additional considerations include proper ventilation, maximum surface temperatures, and thermal controls to prevent overheating under both normal and fault conditions.

The standard addresses mechanical hazards through:

- **Sharp edges and corners:** Evaluation of user-accessible areas
- **Moving parts:** Proper guarding and interlocks
- **Stability requirements:** Prevention of tip-over hazards
- **Enclosure strength:** Resistance to impact and mechanical stress
- **Restricted access areas:** Special requirements for large equipment installations

## Special Requirements and Emerging Technologies

Manufacturers must consider additional safety measures for emerging technologies and specialized applications in accordance with the latest version of IEC 62368-1.

### Battery Safety

For products containing coin or button cell batteries (≤32mm diameter) accessible to children, the latest version of IEC 62368-1 requires compliance with the following, including applicable provisions from Annex M for alternative battery types:

- Battery compartment secured with a tool (e.g., screwdriver or coin)
- Two independent simultaneous movements required to open by hand
- Clear warning markings about ingestion hazards

### Wireless Power Transmission

Objects heated by near-field power transmission include: hair dryers, measuring temperature probes at ambient, 5mm and 2mm distances

### Outdoor Equipment



- Operating temperature range: -33°C to +40°C minimum
- Residual current devices (RCDs) for outdoor socket-outlets (≤30mA)
- Enhanced enclosure requirements per Annex Y or IEC 60529
- Protection against weather and environmental factors



## Global Transition Timeline and Market Requirements

As IEC 62368-1 evolves to Edition 4, understanding regional adoption, legacy standard withdrawals, and new compliance requirements is essential for manufacturers.

### Current Status by Region

| Region         | Standard Adopted            | Withdrawal of Legacy Standards | Notes                          |
|----------------|-----------------------------|--------------------------------|--------------------------------|
| United States  | UL 62368-1:2019 (Edition 3) | December 2020                  | Edition 4, effective July 2025 |
| European Union | EN IEC 62368-1:2020         | December 2020                  | Edition 4 expected late 2025   |
| United Kingdom | BS EN IEC 62368-1:2020      | December 2020                  | Follows EU harmonization       |
| Japan          | JIS C 62368-1               | Adopted                        | Aligned with IEC editions      |

We use cookies to improve your experience on our site. By using our site, you consent to cookies. [Preferences](#) [Reject](#) [Accept All](#) [WPConsent](#)

### Edition 4 Key Changes

The fourth edition, published in 2023, introduces significant changes:

- **End of legacy component acceptance:** Components certified to IEC 60950-1 and IEC 60065 no longer accepted
- **Enhanced battery requirements:** Expanded coverage of lithium battery safety
- **Software-controlled safeguards:** New requirements for software-based safety functions
- **Optical radiation updates:** Modified limits for projectors and LED systems

Manufacturers must plan proactively for Edition 4 compliance as the Date of Withdrawal for Edition 3 approaches in July 2024.



## Testing and Certification Process

To ensure compliance and reduce time-to-market, manufacturers should follow a standardized testing and certification process aligned with global IEC 62368-1 requirements.

### Required Testing Categories

We use cookies to improve your experience on our site. By using our site, you consent to cookies. [Preferences](#) [Reject](#) [Accept All](#) [WPConsent](#)

1. **Electrical Testing:** Dielectric strength, insulation resistance, protective earth continuity
2. **EMC Testing:** Although separate standards apply, safety-related EMC aspects must be addressed
3. **Thermal Testing:** Temperature rise under normal and fault conditions
4. **Flammability Testing:** Materials evaluation via glow wire, needle flame, or other applicable tests
5. **Mechanical Testing:** Impact, stability, and strength tests
6. **Environmental Testing:** Humidity, temperature cycling (for outdoor equipment)

### Certification Pathways

Manufacturers seeking global market access need certification from recognized testing laboratories. Testing laboratories accredited to ISO/IEC 17025 and certification bodies accredited to ISO/IEC 17065 can provide reports and certifications accepted worldwide.

For manufacturers requiring testing across multiple markets, working with laboratories that hold direct certification authority or recognition under Mutual Recognition Agreements (MRAs) can significantly reduce time-to-market and testing costs. Real-time test tracking and automated reporting systems further streamline the certification process.



## Compliance Best Practices for Manufacturers

Manufacturers can achieve faster and more reliable IEC 62368-1 compliance by integrating hazard-based safety engineering early and maintaining thorough documentation.

### Early Design Integration

We use cookies to improve your experience on our site. By using our site, you consent to cookies. [Preferences](#) [Reject](#) [Accept All](#) [WPConsent](#)

- Reduces redesign costs and delays
- Enables optimization of safeguard strategies
- Allows for innovative safety solutions not possible under prescriptive standards
- Facilitates faster certification cycles

### Documentation Requirements

Maintain comprehensive technical documentation, including:

- Safeguard analysis and effectiveness verification
- Bill of materials with component certifications (Edition 3 and earlier)
- Test reports from accredited laboratories
- Risk assessments (where applicable)
- User instructions and warnings

## Managing Component Obsolescence

**With Edition 4 eliminating acceptance of legacy component certifications, manufacturers must:**

- Audit existing products for components certified only to IEC 60950-1 or IEC 60065
- Identify replacement components certified to IEC 62368-1
- Plan product recertification timelines
- Maintain documentation flexibility for component changes

We use cookies to improve your experience on our site. By using our site, you consent to cookies.

Preferences

Reject

Accept All

WPConsent

IEC 62368-1 sets the future standard for technology safety, combining a hazard-based approach with user protection and design flexibility. Understanding energy classifications, safeguards, and HBSE is key to compliance.

As global markets move to Edition 4 and phase out legacy components, manufacturers must act now. Partnering with experienced, accredited laboratories offering comprehensive testing, real-time tracking, and global certification support can simplify and accelerate compliance.

## Ready to Navigate IEC 62368-1 Compliance?

For manufacturers requiring comprehensive safety testing and certification services, MiCOM Labs provides A2LA-accredited safety testing capabilities, along with RF/wireless testing, EMC testing, and global market access services. Our ISO/IEC 17025 and 17065 accreditations ensure your testing meets international requirements, while our proprietary platforms enable real-time project tracking and faster certification timelines. [Contact us today](#) to learn more about IEC 62368-1 compliance.



Enter Your Email



### About Us

Timeline

US Headquarters

China Office

India Office

Certificate Validation

### Certification

Product Certifications

Global Market Access

MiPassport®

MiCybercert

### Contact Us

575 Boulder Court, Pleasanton, California, 94566, USA

+1 (925) 462-0304

We use cookies to improve your experience on our site. By using our site, you consent to cookies.

Preferences

Reject

Accept All

WPConsent